# Challenge 3: Impact Statement Card

Anticipating the societal impacts of AI systems is a critical phase of project planning. Evaluate how your solution will affect various communities and stakeholders, both positively and negatively.

## Intended Positive Impacts

*What are the direct benefits of your solution? How does it empower or assist the target audience?*

- Increased crop yields through timely, climate-aware planting advice.
- Enhanced economic stability and food security for smallholder families.
- Optimized use of resources (water, fertilizer), promoting sustainable farming.

## Potential Negative Consequences

*What could go wrong? Consider misuse, systemic bias, job displacement, or environmental cost.*

- Severe financial loss if the model provides inaccurate weather or crop advice.
- Erosion of traditional ecological knowledge if farmers become over-reliant on the AI.
- Exclusion of farmers who do not speak the supported languages (English/Swahili).

## Mitigation Strategies

*For each negative consequence identified above, describe the guardrails, technical safeguards, or policy steps your team will implement to minimize harm.*

**1. Accuracy/Financial Loss:** The model will include confidence thresholds; if certainty is low, it will explicitly advise consulting a human extension worker. We will not provide definitive commands, only probabilities and suggestions.

**2. Knowledge Erosion:** The UI will frame advice as a supplement to, rather than a replacement for, traditional practices, encouraging shared community decision-making.

**3. Language Barriers:** Phase 2 of our roadmap includes fine-tuning the SLM on local dialects using community-sourced voice data to expand accessibility.